Update
Mason 2YP

November 1, 2026

Contents

1 Empirical demand **2**

1.1 Notation . . . 2
1.2 Price equation and control residuals . . . 2
1.3 Utility, heterogeneity, and choice probabilities . . . 3
1.4 Unaggregated flavor concentration . . . 3
1.5 Summary statistics . . . 6
1.6 Results . . . 7
1.7 Discussion of Results . . . 9

2 Supply: two-period pricing **9**

2.1 Notation . . . 9
2.2 Utilities and shares . . . 10
2.3 Terminal-period optimization . . . 10
2.4 First-period optimization . . . 11
2.5 Profit derivative . . . 11
2.6 Conditional derivative signs . . . 11
2.7 Price ratios and conditional profit for all histories . . . 11
2.8 Numerical specification . . . 12
2.9 Discussion of Supply . . . 16

1 Empirical demand

1.1 Notation

Symbol	Definition
i	Household.
$t \in \mathcal{T}_i$	Retained household purchase week, ending Saturday.
$\mathcal{H}_i$	Observed purchase weeks used to construct history, including weeks subsequently excluded from the likelihood.
$j, k \in \mathcal{J}$	Flavor categories; $\mathcal{J} = \{\text{other}, \text{berry}, \text{plain}\}$.
0	Outside alternative, included in the reported fit.
Q_{ijt}, Q_{it}	Quantity purchased of flavor j; total inside quantity $Q_{it} = \sum_j Q_{ijt}$.
x_{ijt}	Inside quantity share of flavor j, Q_{ijt}/Q_{it}.
S_{ijt}, N_{ijt}	Cumulative own-flavor share stock and immediately preceding calendar week's other-flavor share.
λ, Ξ_{ijt}	Fixed history weight and combined history index; $\lambda = 0.7$.
p_{ijt}	Exposure-store flavor price.
v_{ijt}, $\widehat{v}_{ijt}$	First-stage price disturbance and estimated residual.
b_0	Common inside intercept relative to the outside alternative.
β_j	Flavor intercept relative to other; $\beta_{\text{other}} = 0$ in the reported fit.
α_i, μ_α, σ_α	Positive price sensitivity and mean/SD of its logarithm. Price enters utility as $-\alpha_i p$.
γ_i, μ_γ, σ_γ	Household history coefficient, common across flavors, and its population mean/SD.
ρ	Common coefficient on the price-control residual.
ε_{ijt}, V_{ijt}, U_{ijt}	Unit-scale type-I extreme-value shock, deterministic utility, and total utility.
P_{ijt}	Conditional choice probability for given household type.

1.2 Price equation and control residuals

Let m index consumer DMAs and m_{it} be household i's DMA in week t. $\mathcal{Q}_{mt}$ contains the distinct store–week–UPC price quotes in DMA m, pooling all flavors. Their count and price sum are

$$n_{mt} = |\mathcal{Q}_{mt}|, \tag{1}$$

$$P_{mt} = \sum_{q \in \mathcal{Q}_{mt}} p_q, \tag{2}$$

The instrument z_{it} is the same-week mean price outside the household's DMA:

$$z_{it} = \frac{\sum_{m \neq m_{it}} P_{mt}}{\sum_{m \neq m_{it}} n_{mt}}, \qquad \sum_{m \neq m_{it}} n_{mt} > 0, \tag{3}$$

$$p_{ijt} = \pi_j z_{it} + \tau_{jt} + \kappa_{j,m_{it}} + v_{ijt}, \qquad (i,t) \in \mathcal{D}_j. \tag{4}$$

Here π_j is the instrument slope and τ_{jt} and κ_{jm} are flavor-specific week and DMA fixed effects. $\mathcal{D}_j$ contains the usable household–weeks for flavor j. The same z_{it} enters each flavor equation. Each household–week has equal first-stage weight. The table reports the one-instrument Wald statistic F_j.

Table 2: Price first stages

Flavor	$\widehat{\pi}_j$	Cluster SE	F_j	Partial R^2	Observations	Households
Other	-34.6795	0.9662	1,288.26	0.0213	252,033	14,965
Berry	-33.1049	1.3767	578.20	0.0192	94,753	11,434
Plain	-28.5850	3.0847	85.87	0.0076	41,835	7,510

1.3 Utility, heterogeneity, and choice probabilities

Fatigue is a decline in utility from a flavor as the household accumulates past purchases of that flavor. *Novelty* is a utility gain from choosing a flavor different from those purchased in the immediately preceding calendar week. Define the corresponding history measures by

$$S_{ijt} = \sum_{\substack{s \in \mathcal{H}_i \\ s<t}} x_{ijs}, \qquad N_{ijt} = \begin{cases} \sum_{k \neq j} x_{ik,t-1}, & t-1 \in \mathcal{H}_i, \\ 0, & t-1 \notin \mathcal{H}_i, \end{cases} \tag{5}$$
$$\Xi_{ijt} = (1-\lambda) N_{ijt} - \lambda S_{ijt}, \qquad \lambda = 0.7.$$

Thus $\gamma_i \Xi_{ijt} = 0.3\gamma_i N_{ijt} - 0.7\gamma_i S_{ijt}$: positive γ_i implies both fatigue and a preference for novelty, while negative γ_i implies reinforcement from accumulated own-flavor purchases and a preference for repeating recent flavors. Fatigue accumulates without depreciation; novelty depends only on the preceding calendar week and is zero after a purchase gap. The fixed weight λ ties the two effects together, so they are not separately estimated.

$$U_{ijt} = V_{ijt} + \varepsilon_{ijt}, \tag{6}$$
$$V_{ijt} = b_0 + \beta_j - \alpha_i p_{ijt} + \gamma_i \Xi_{ijt} + \rho \widehat{v}_{ijt}, \tag{7}$$
$$\beta_{\text{other}} = 0, \tag{8}$$

The household type components $Z_{i\alpha}$ and $Z_{i\gamma}$ are independent standard normals, fixed over each household's history:

$$\log \alpha_i = \mu_\alpha + \sigma_\alpha Z_{i\alpha}, \qquad \alpha_i > 0, \tag{9}$$
$$\gamma_i = \mu_\gamma + \sigma_\gamma Z_{i\gamma}, \tag{10}$$
$$\begin{pmatrix} Z_{i\alpha} \\ Z_{i\gamma} \end{pmatrix} \sim N(\mathbf{0}, I_2), \tag{11}$$

Let $\mathcal{A}_{it}$ contain the inside flavors with finite positive prices at the household's exposure store. Conditional on household type,

$$P_{ijt}(\alpha_i, \gamma_i) = \frac{\exp(V_{ijt})}{1 + \sum_{k \in \mathcal{A}_{it}} \exp(V_{ikt})}, \qquad j \in \mathcal{A}_{it}, \tag{12}$$

1.4 Unaggregated flavor concentration

Let $\mathcal{F}$ be the detailed flavor labels and Q_{ift} the quantity of flavor f bought by household i in week t. Household full-history quantity shares w_{if} and flavor HHI H_i are

$$w_{if} = \frac{\sum_{t \in \mathcal{H}_i} Q_{ift}}{\sum_{g \in \mathcal{F}} \sum_{t \in \mathcal{H}_i} Q_{igt}}, \tag{13}$$

$$\sum_{f \in \mathcal{F}} w_{if} = 1, \tag{14}$$

$$H_i = \sum_{f \in \mathcal{F}} w_{if}^2. \tag{15}$$

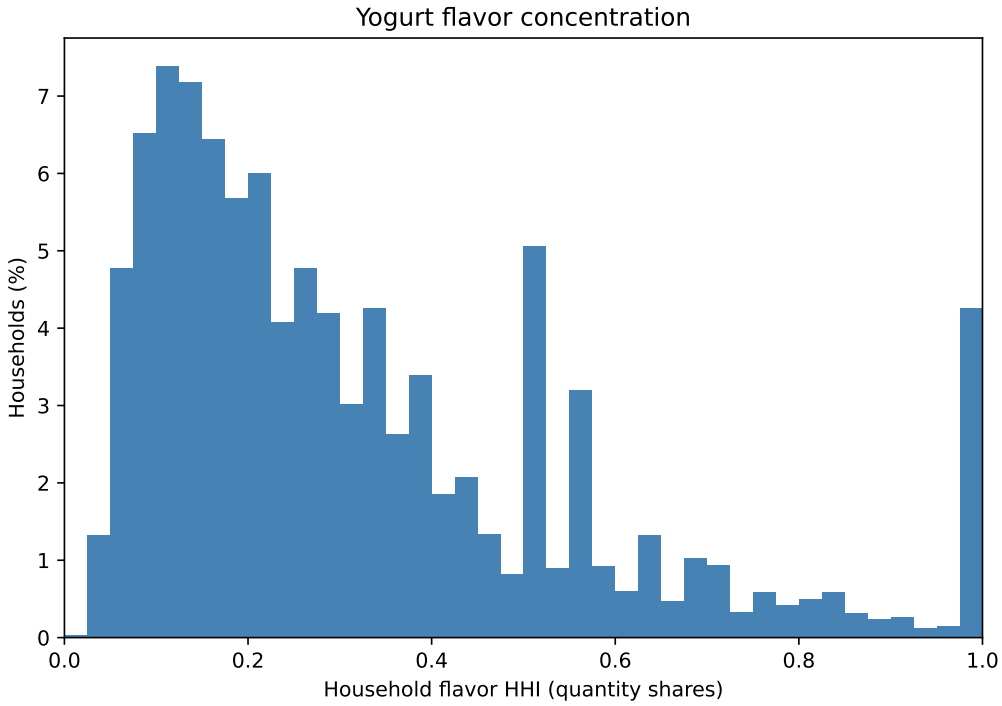


Figure 1: Household flavor HHI; unaggregated flavor labels, shares of quantity purchased.

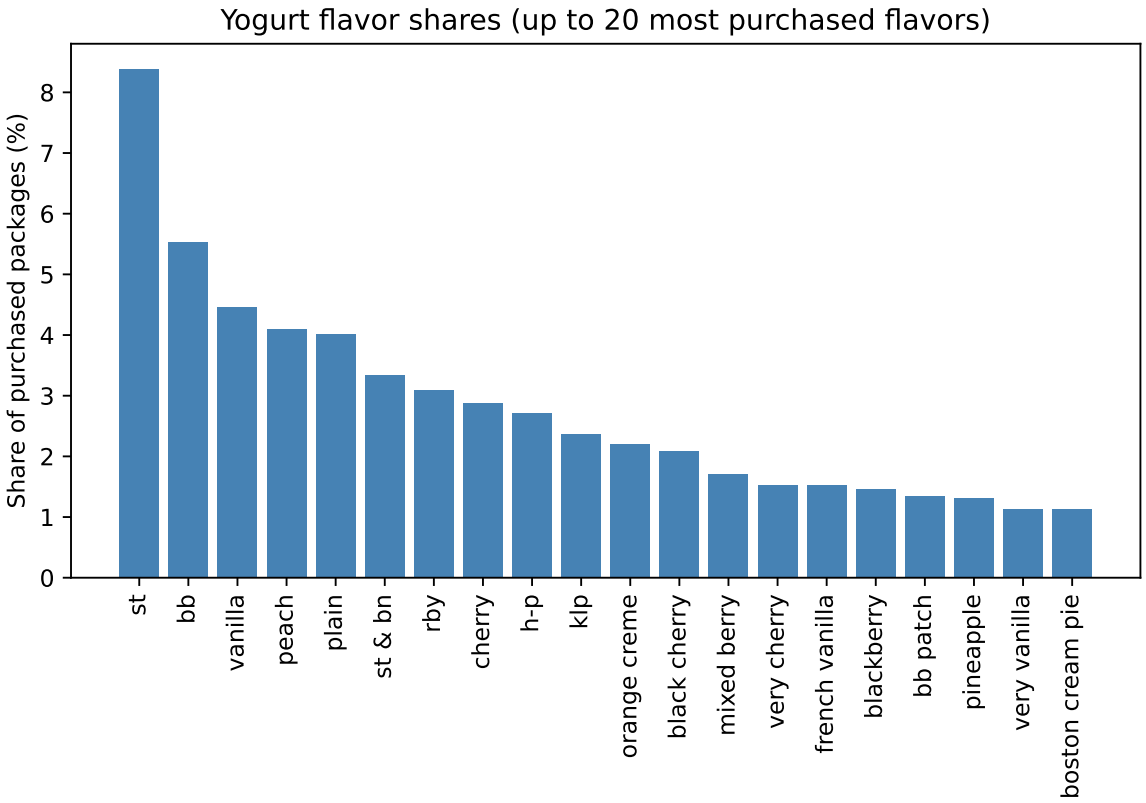


Figure 2: Shares of quantity purchased; 20 most purchased unaggregated flavor labels.

1.5 Summary statistics

The descriptive sample covers 2006–2015. Trips, UPCs, and quantity purchased in the matched yogurt column refer to yogurt only.

Table 3: Household purchase summaries

Statistic	Full panel	Matched yogurt
Households	37,827	15,106
Mean observed panel years	3.96	3.16
Mean trips per household	493.13	20.26
Mean trips per approximate observed week	2.27	0.11
Median trips per approximate observed week	1.93	0.07
Mean distinct UPCs per household	1178.31	13.89
Mean quantity purchased per household	4669.07	93.77
One-person households (%)	100.00	100.00
Two-person households (%)	0.00	0.00
Hispanic origin (%)	3.10	2.59

Table 4: Yogurt purchase dynamics: detailed flavors and estimation categories

Statistic	Detailed flavors	Three categories
Households	15,106	15,106
Observed purchase household–weeks	282,648	282,648
Mean flavors/categories bought per household	10.76	1.93
Mean household flavor HHI	0.32	0.78
Mixed-flavor/category weeks (%)	37.88	17.45
Weeks with quantity purchased above seven (%)	21.72	21.72
Eligible adjacent single-flavor/category week pairs	30,392	48,166
Switch rate among eligible pairs (%)	46.75	12.85
Mean single-flavor/category spell length (weeks)	1.10	1.22
Median gap between purchase weeks (weeks)	3.00	3.00

An eligible transition requires purchases in consecutive calendar weeks and exactly one flavor (or category) in each week. Mixed baskets and gaps are excluded from the switch-rate denominator, thus it is illustrative only. Spells count consecutive calendar weeks with the same single flavor or category; a gap breaks the spell. HHI uses each household's full-history shares of quantity purchased.

Table 5: Quantity shares and transitions for the three estimation categories

	Quantity share (%)	Next category (row probabilities, %)		
Category		Other	Berry	Plain
Other	78.37	92.25	5.79	1.96
Berry	14.60	45.27	52.93	1.79
Plain	7.03	18.09	1.85	80.06

1.6 Results

Table 6: Estimation sample and settings; single-person households

Setting	Value
Years	2006–2015
Households	15,105
Household–weeks	281,695
Capacity setting	7
λ	0.7
Integration draws	32
Optimization starts	1
Household size	1
Converged	Yes
Parameters at bounds	0
Negative log likelihood	1,642,195.0310

Table 7: Demand estimates

Parameter	Estimate	Standard error	95% normal interval
b_0	1.3486	0.0066	[1.3356, 1.3616]
β_{berry}	-0.6504	0.0032	[-0.6567, -0.6440]
β_{plain}	-0.1040	0.0063	[-0.1164, -0.0916]
μ_α	-0.3620	0.0068	[-0.3754, -0.3486]
μ_γ	-0.0588	0.0002	[-0.0593, -0.0583]
σ_α	0.6654	0.0045	[0.6567, 0.6742]
σ_γ	0.0634	0.0003	[0.0629, 0.0639]
ρ	0.1675	0.0044	[0.1588, 0.1762]

Table 8: Implied willingness to pay

Utility coefficient	Mean WTP	Population SD
Inside constant b_0	2.4170	1.8040
Berry vs. other: β_{berry}	-1.1656	0.8700
Plain vs. other: β_{plain}	-0.1864	0.1391
History index: γ_i	-0.1054	0.1622
Novelty: $(1-\lambda)\gamma_i$	-0.0316	0.0487
Own-flavor stock: $-\lambda\gamma_i$	0.0738	0.1135

WTP is in the same monetary units as price, per unit of the indicated regressor. Flavor coefficients are relative to other; b_0 is other versus the outside option at zero history and control residual. The last two rows hold the other history regressor fixed, with $\lambda = 0.7$.

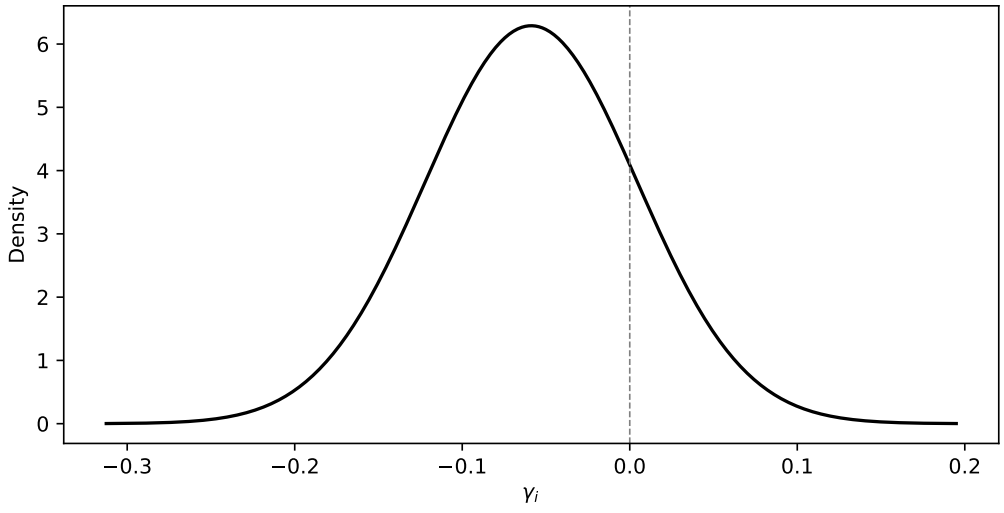


Figure 3: Fitted household history coefficient: $\gamma_i \sim N(-0.0588034,\, 0.0634273^2)$.

1.7 Discussion of Results

I have 15,105 single-person households in my estimation sample, a 30x increase over the past runs. The descriptive yogurt sample has 15,106 households, compared with 15,105 retained for estimation after the usable-week restrictions. In the descriptive yogurt sample, households are observed for an average of about three years and make 20 yogurt-only trips, on average. The full panel covers 37,827 households and averages 493 trips across all products, so that is a different household set. In an approximate observed week, there are only about 0.1 yogurt trips. Households in the descriptive yogurt sample buy about 14 distinct yogurt UPCs as well as about 94 total units of yogurt. Switching stats show that there are nearly 11 flavors bought in the disaggregated sample and about two in the three-flavor category aggregation. Mean within-household flavor HHI in the disaggregated is 0.32 versus 0.78 in the aggregated, expected given how large the categories can be. Similarly, we have more than a third of weeks with a mixed basket in the disaggregated sample versus about 17% in the three category specification. Switching rate is higher with more flavors to switch between, again to be expected. I think using the disaggregated sample for illustration then three category to ease estimation makes sense. Appreciate thoughts here.

Looking at the within household flavor concentration plot shows that most consumers consume many flavors of yogurt. We do see a large grouping at the single flavor point, but they still only account for about 3% of households. Interestingly, there is another large cluster around the 50% point, which is good evidence of switching behavior varying across households. Strawberry yogurt is the most purchased flavor at about 8.5% of all sales. Notably, nine berry flavors populate the top 20 flavors, showing that it is a good grouping. I do think aggregating "other" into "other fruits", "dessert flavors", and "other" may be good once the three category specification is working as we want it to.

Estimation has taken a massive improvement with the changes to the novelty versus fatigue mechanism in Ξ alongside Josh's proposed changes to changing $\beta_{ij} \to \beta_j$ and $\gamma_{ij} \to \gamma_i$. γ is now identified with precision, though the effect and spread are quite small. Price sensitivity has a large spread and large negative effect. Looking at the distribution, assumed to be normal, we do see some agents prefer novelty and show fatigue, while most show reinforcement from same-flavor accumulation and prefer repetition in consumption of their preferred products. From the implied willingness to pay, relative to other flavors, berry and plain yogurt would require subsidization for consumers to value the flavors equivalently. For history, we can disaggregate γ into the novelty and fatigue terms. Novelty leads to a loss of value of \$-0.03 on average, implying an additional unit (addition to flavor share) reduces the value of a unit. For persistence, about \$0.07 in WTP is added per-unit on average for same flavor accumulation. Both effects hold the other fixed. Overall, this has been a large improvement. I think a lot is due to the sample size growth and then the estimation procedure changes proposed by Josh.

2 Supply: two-period pricing

2.1 Notation

Symbol	Definition
$t \in \{1, 2\}$, j, 0	Period, inside product, and outside alternative.
$x_0 = h$, $x_1 = a$	Initial realized choice and realized first-period choice; each belongs to $\{j, 0\}$.

Symbol	Definition
β, $\alpha > 0$, γ	Fixed, firm-observed type's inside intercept, positive price sensitivity, and symmetric switching coefficient. Type indices are suppressed.
c, $\delta \in (0,1)$	Marginal cost and discount factor.
$p_t(h)$	Price conditional on preceding realized choice h. Prices belong to $\mathbb{R}$.
$q(h)$	Signed history indicator: $q(j) = -1$, $q(0) = 1$.
$s_t(h)$, $s_{0t}(h)$	Inside and outside choice probabilities.
$W_2(a)$	Optimized terminal profit conditional on $x_1 = a$.
$\Pi(p_1; h)$, $\Pi^*(h)$	Expected discounted profit at candidate first-period price and at its optimum.

2.2 Utilities and shares

V_{jt}^{raw} and V_{0t}^{raw} are deterministic utilities before subtracting the outside utility. Their difference is $v_t(h)$:

$$V_{jt}^{\text{raw}} = \beta - \alpha p_t + \gamma \mathbf{1}\{x_{t-1} = 0\}, \tag{16}$$

$$V_{0t}^{\text{raw}} = \gamma \mathbf{1}\{x_{t-1} = j\}, \tag{17}$$

$$\begin{aligned} v_t(h) &= V_{jt}^{\text{raw}} - V_{0t}^{\text{raw}} \\ &= \beta - \alpha p_t(h) + \gamma[\mathbf{1}\{h = 0\} - \mathbf{1}\{h = j\}] \qquad (18) \\ &= \beta - \alpha p_t(h) + q(h)\gamma, \end{aligned}$$

$$s_t(h) = \frac{e^{v_t(h)}}{1 + e^{v_t(h)}}, \qquad s_{0t}(h) = 1 - s_t(h). \tag{19}$$

$$\frac{\partial s_t}{\partial p_t} = -\alpha s_t(1 - s_t), \tag{20}$$

$$\left.\frac{\partial s_t}{\partial \gamma}\right|_{p_t,h} = q(h) s_t(1 - s_t), \tag{21}$$

2.3 Terminal-period optimization

A star marks an optimized price or its associated choice probability. For each $a \in \{j, 0\}$:

$$W_2(a) = \max_{p_2(a) \in \mathbb{R}} s_2(a)[p_2(a) - c], \tag{22}$$

$$\begin{aligned} 0 &= \frac{\partial}{\partial p_2}\{s_2(p_2 - c)\} \\ &= s_2 + (p_2 - c)\frac{\partial s_2}{\partial p_2} \qquad (23) \\ &= s_2 - \alpha s_2(1 - s_2)(p_2 - c), \end{aligned}$$

$$1 = \alpha(1 - s_2^*)(p_2^* - c), \tag{24}$$

$$p_2^*(a) = c + \frac{1}{\alpha[1 - s_2^*(a)]}, \tag{25}$$

$$W_2(a) = \frac{s_2^*(a)}{\alpha[1 - s_2^*(a)]}. \tag{26}$$

2.4 First-period optimization

$$\begin{aligned}\Pi(p_1;h) &= s_1(h)(p_1-c)+\delta\{s_1(h)W_2(j)+[1-s_1(h)]W_2(0)\}\\ &= \delta W_2(0)+s_1(h)[p_1-c+\delta[W_2(j)-W_2(0)]],\end{aligned} \tag{27}$$

$$\begin{aligned}\partial_{p_1}\Pi &= s_1+\partial_{p_1}s_1[p_1-c+\delta[W_2(j)-W_2(0)]]\\ &= s_1-\alpha s_1(1-s_1)[p_1-c+\delta[W_2(j)-W_2(0)]],\end{aligned} \tag{28}$$

$$0 = 1-\alpha(1-s_1^*)[p_1^*-c+\delta[W_2(j)-W_2(0)]], \tag{29}$$

$$p_1^*(h) = c+\frac{1}{\alpha[1-s_1^*(h)]}-\delta[W_2(j)-W_2(0)], \tag{30}$$

$$\Pi^*(h) = \delta W_2(0)+\frac{s_1^*(h)}{\alpha[1-s_1^*(h)]}. \tag{31}$$

2.5 Profit derivative

$$\begin{aligned}\frac{\mathrm{d}\Pi^*(h)}{\mathrm{d}\gamma} &= \partial_\gamma\Pi(p_1;h)\big|_{p_1=p_1^*(h)}\\ &= \frac{q(h)s_1^*(h)-\delta s_1^*(h)s_2^*(j)+\delta[1-s_1^*(h)]s_2^*(0)}{\alpha}.\end{aligned} \tag{32}$$

2.6 Conditional derivative signs

$$\frac{\mathrm{d}\Pi^*(0)}{\mathrm{d}\gamma} = \frac{s_1^*(0)[1-\delta s_2^*(j)]+\delta[1-s_1^*(0)]s_2^*(0)}{\alpha} > 0, \tag{33}$$

$$\frac{\mathrm{d}\Pi^*(j)}{\mathrm{d}\gamma} > 0 \iff \delta[1-s_1^*(j)]s_2^*(0) > s_1^*(j)[1+\delta s_2^*(j)]. \tag{34}$$

2.7 Price ratios and conditional profit for all histories

For each history $(h,a)=(x_0,x_1)\in\{(0,0),(0,j),(j,j),(j,0)\}$,

$$R_{h,a} = \frac{p_1^*(h)}{p_2^*(a)}. \tag{35}$$

Each ratio requires a nonzero denominator. A ratio below one means the second-period price exceeds the first-period price; a ratio above one means the first-period price exceeds the second-period price.

Conditional on the realized first-period choice a, profit is

$$\Pi^*(h,a) = \mathbf{1}\{a=j\}[p_1^*(h)-c]+\delta W_2(a). \tag{36}$$

The first-period margin is realized, while terminal profit remains expected. Fully realized two-period profit would also condition on x_2. Averaging over first-period choices recovers optimal expected profit:

$$\Pi^*(h) = s_1^*(h)\Pi^*(h,j)+[1-s_1^*(h)]\Pi^*(h,0). \tag{37}$$

2.8 Numerical specification

$\beta = 2, \qquad \alpha = 1, \qquad \delta = 0.92, \qquad c = 0, \qquad \gamma \in [-3, 3]$ (601 grid points per initial state).

Table 10: Selected supply solutions

h	γ	$p_1^*(h)$	$p_2^*(j)$	$p_2^*(0)$	$\Pi^*(h)$	$\mathrm{d}\Pi^*(h)/\mathrm{d}\gamma$
j	-3	3.395	3.926	1.120	5.087	-1.387
j	-1	2.236	2.557	1.567	2.669	-0.959
j	0	2.000	2.000	2.000	1.920	-0.500
j	1	2.209	1.567	2.557	1.731	0.125
j	3	3.592	1.120	3.926	2.702	0.668
0	-3	-0.780	3.926	1.120	0.913	0.195
0	-1	1.045	2.557	1.567	1.478	0.385
0	0	2.000	2.000	2.000	1.920	0.500
0	1	2.956	1.567	2.557	2.478	0.615
0	3	4.801	1.120	3.926	3.912	0.804

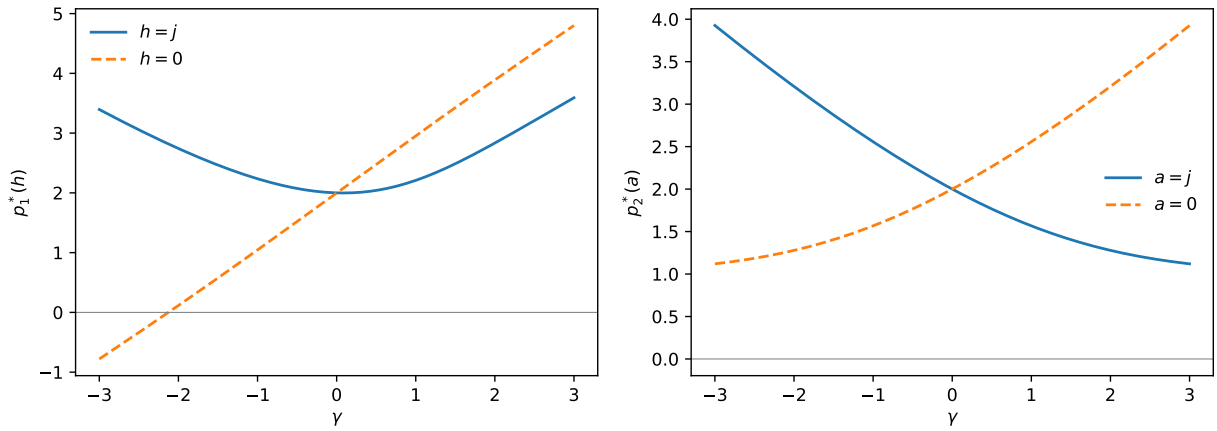


Figure 4: $p_1^*(h)$ and $p_2^*(a)$; $\beta = 2$, $\alpha = 1$, $\delta = 0.92$, $c = 0$.

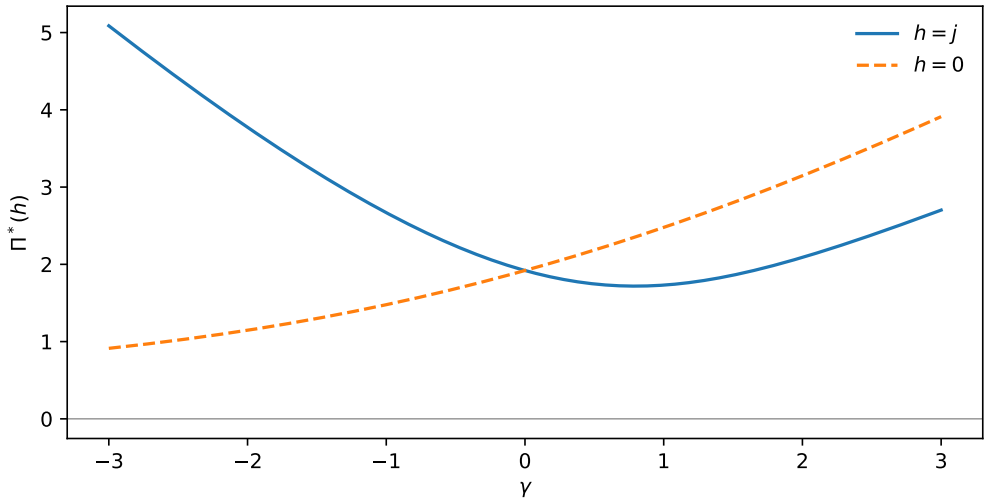


Figure 5: $\Pi^*(h)$; $\beta = 2$, $\alpha = 1$, $\delta = 0.92$, $c = 0$.

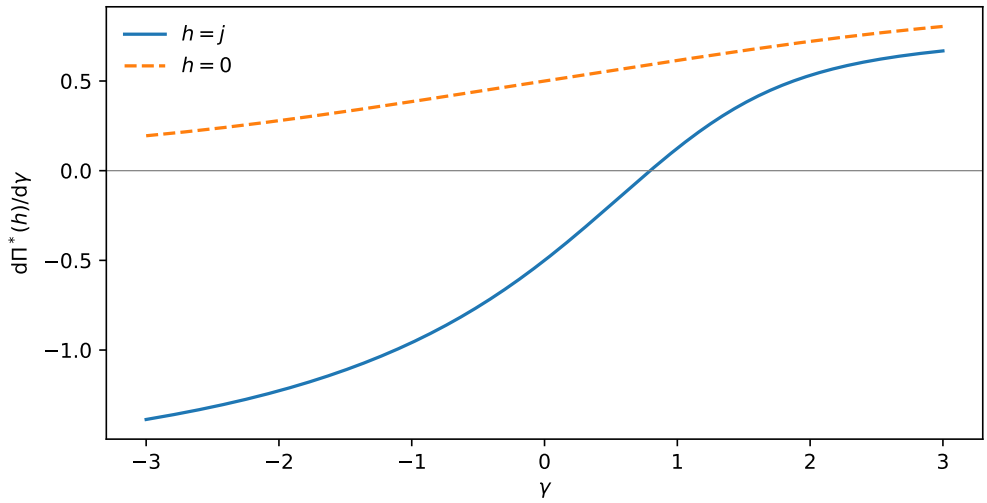


Figure 6: $\mathrm{d}\Pi^*(h)/\mathrm{d}\gamma$; $\beta = 2$, $\alpha = 1$, $\delta = 0.92$, $c = 0$.

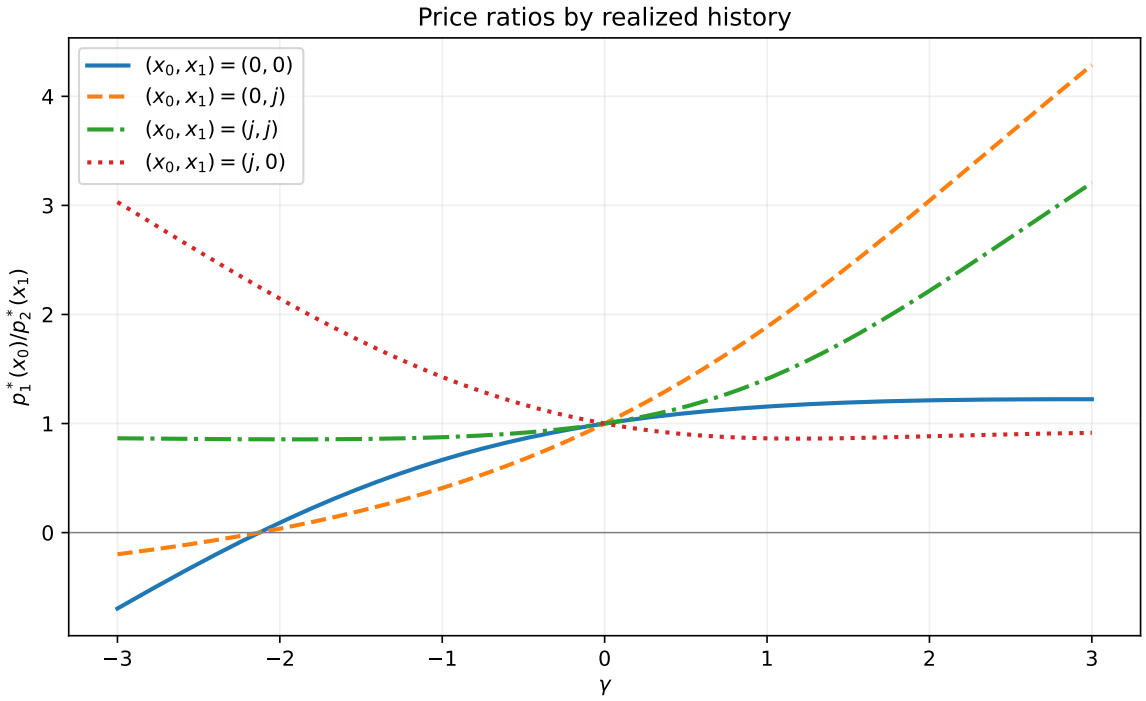


Figure 7: Price ratios $R_{h,a} = p_1^*(h)/p_2^*(a)$ for all four histories $(h, a) = (x_0, x_1)$; $\beta = 2$, $\alpha = 1$, $\delta = 0.92$, $c = 0$.

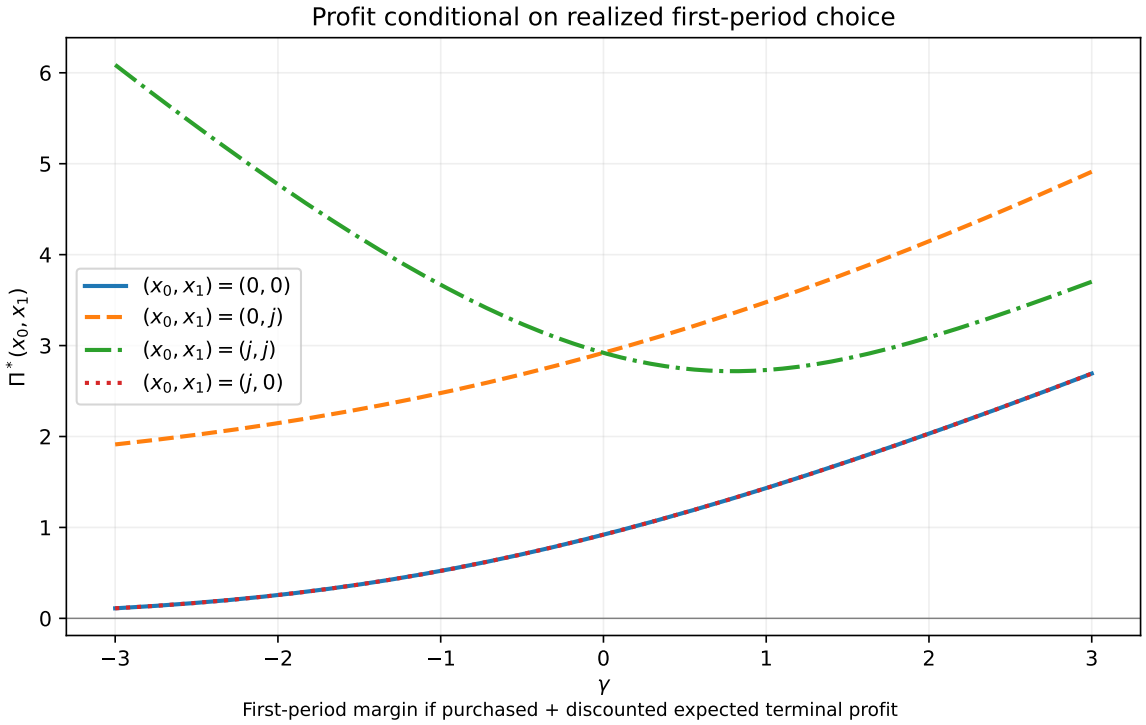


Figure 8: Conditional profit $\Pi^*(h, a)$ for all four histories. First-period profit is realized and terminal profit is expected. The $(0, 0)$ and $(j, 0)$ curves coincide; $\beta = 2$, $\alpha = 1$, $\delta = 0.92$, $c = 0$.

2.9 Discussion of Supply

Optimal first-period prices are U-shaped for consumers initially inside ($h = j$), while prices for consumers initially outside ($h = 0$) increase with γ. Terminal prices depend on the most recent choice: positive γ raises willingness to buy after $a = 0$ and lowers willingness to buy after $a = j$.

For negative γ, persistence supports an induce-then-harvest strategy: the firm can offer an initially outside consumer a low or even negative first-period price, then charge a higher terminal price after purchase. The price ratio $R_{0,j}$ captures this pattern.

For high positive γ, the timing reverses. Initially outside consumers want to switch into j, allowing the firm to charge a high first-period price. After purchase, these same consumers want to switch away, so the firm lowers the terminal price to offset some of the loss of demand. This retention discount maximizes terminal profit but does not prevent purchase shares from falling as switching preferences strengthen.

For initially inside consumers, allowing a first-period departure also has value: choosing 0 makes the consumer more willing to return in period 2. When $W_2(0) > W_2(j)$, the first-period pricing condition includes a positive opportunity cost $\delta[W_2(0) - W_2(j)]$. The firm requires a larger current margin to compensate for placing a buyer in the less profitable continuation state. At sufficiently high γ, first-period prices rise even while the first-period purchase probability falls.

The conditional-profit curves distinguish margins from the probabilities of earning them. Both $(0, j)$ and (j, j) receive the same terminal retention price and continuation value; their profit difference comes entirely from first-period margins. The rising (j, j) conditional-profit curve at high γ describes a large margin on an increasingly rare first-period purchase. For $(0, 0)$ and $(j, 0)$, no first-period margin is earned, so both conditional profits equal $\delta W_2(0)$. Optimal expected profit weights each conditional profit by its first-period choice probability.

These simulations use an illustrative calibration. A future step is to integrate the estimated preference distribution into the pricing exercise and compare the resulting prices with observed prices.